

# FINAL EXPLANATION: GENERAL SCIENCE GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

## FINAL EXPLANATION: GENERAL SCIENCE GRADE 10 – SUB-STRAND 3.3 WAVES – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

### INSTRUCTIONS GIVEN TO STUDENTS

Driving question: How do waves carry energy through our environment, and how can understanding their behaviour protect Kenyan communities, improve technology, and help us make sense of the world around us? In this task you will act as a scientist explaining the Lake Victoria fisherman's puzzle. Write your answers in the space under each prompt. Use the vocabulary and ideas from Lessons 1 to 8: wavelength, frequency, wave speed, reflection, refraction, diffraction, and the wave equation. Show every step of any calculation, including units. Refer back to the fisherman near Kisumu in each answer. Write in full sentences, and label any diagram you draw.

### Part 1: The wave equation and the fisherman's sounds and waves

**A fisherman near Kisumu hears a deep thunder rumble and sees water waves crossing the lake.**

**(a) Write the wave equation and say what each symbol means. (b) A Lake Victoria boat-engine wave has wavelength 0.8 m and speed 3.2 m/s. Calculate its frequency and its period. (c) Explain why, in the same medium, a wave of higher frequency has a shorter wavelength. (d) Say how the wave equation helps you describe the low-frequency thunder rumble compared with a high-pitched sound.**

(a)  $v = f\lambda$ , where  $v$  is wave speed (m/s),  $f$  is frequency (Hz) and  $\lambda$  is wavelength (m). (b)  $f = v/\lambda = 3.2 \div 0.8 = 4$  Hz. Period  $T = 1/f = 1 \div 4 = 0.25$  s. (c) In a given medium the wave speed  $v$  stays constant. Since  $v = f\lambda$ , if  $f$  increases then  $\lambda$  must decrease so the product stays the same, so frequency and wavelength are inversely proportional. In the slinky demonstration, moving the hand faster gave more crests and shorter wavelengths while the wave speed along the spring stayed the same. (d) Rearranging gives  $\lambda = v/f$ . Sound travels at about the same speed in air, so the deep, low-frequency thunder rumble has a long wavelength and a high-pitched sound has a short wavelength. This matters for the fisherman because, as shown later, long wavelengths bend around obstacles like the headland more easily.

### Part 2: Reflection – what the echo can and cannot explain

**In Lesson 3 you clapped at measured distances from the school perimeter wall and timed**

(a) The angle of incidence equals the angle of reflection. (b) An echo is a sound wave that has reflected off a hard surface, such as the wall or the rocky cliff, and returned to the listener. (c) The sound travels to the surface and back, so the time  $t$  covers twice the distance  $d$ . Dividing by 2 gives the one-way distance. (d)  $d = (1500 \times 0.2) / 2 = 300 / 2 = 150$  m. (e) Reflection can explain why the fisherman hears some thunder coming from the direction of the rocky cliff: the hard rock reflects some of the sound energy towards

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| <p>the echo.</p> <p>(a) State the law of reflection. (b) Explain what an echo is. (c) The echo-distance formula is <math>d = (v \times t) / 2</math>. Explain why the time is divided by 2. (d) A sonar pulse from a boat in Kisumu harbour travels at 1500 m/s and returns after 0.2 s. Calculate the depth of the water. (e) Reflection can explain part of what the fisherman hears. Explain what it can explain and what it cannot explain about the sound reaching the cove behind the headland.</p> | <p>him. It cannot explain how sound reaches the shadowed cove, because reflection needs a path from the source to a surface facing the storm, then onward to the listener. There is no reflecting surface facing the storm that sends sound into the cove, so another wave behaviour is needed.</p> |
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### Part 3: Refraction of the water waves

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| <p><b>The water waves from the storm change direction as they cross from the deep open lake into the shallow bay near the Kisumu shore.</b></p> <p>(a) Define refraction. (b) Explain why the wavefronts change direction at the boundary between deep and shallow water, and say which way they bend relative to the normal. (c) A wave has frequency 680 Hz. It travels at 340 m/s in one region of air and slows to 320 m/s in another. Calculate the wavelength in each region. (d) What stays the same and what changes for a wave crossing a boundary? (e) Describe one Kenyan example of refraction of sound that you studied.</p> | <p>(a) Refraction is the change in direction of a wave caused by a change in its speed when it crosses a boundary between two media. (b) In the shallow bay the water waves slow down. Part of each wavefront slows before the rest, so the wavefront swings round and the wave bends towards the normal. This is the same behaviour seen in the Mombasa coastal swell video. (c) <math>\lambda = v/f</math>. At 340 m/s: <math>340 \div 680 = 0.50</math> m. At 320 m/s: <math>320 \div 680 \approx 0.47</math> m. (d) Frequency stays the same at the boundary. Speed changes, and wavelength changes in proportion to speed (<math>v = f\lambda</math>), so the wavelength is shorter in the slower region. (e) At night a temperature inversion makes sound travel faster in the upper warm layer, so it refracts back downward. This is why Lamu fishermen hear voices carrying farther across the water. Refraction explains the change in direction of the water waves but not sound reaching the cove behind a solid headland.</p> |
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### Part 4: Diffraction and putting the whole phenomenon together

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| (a) Define diffraction and say | (a) Diffraction is the spreading of waves around obstacles and through gaps. It is greatest when the gap or obstacle size is close to |
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| <p>when it is greatest. (b) Use diffraction to explain how the deep thunder rumble reaches the fisherman even when the storm is behind the headland. (c) The mobile-phone signal fades when you go behind a hill in the Rift Valley. Explain what is happening and what engineers do about it. (d) Write a short explanation of the full fisherman phenomenon, naming which wave behaviour (reflection, refraction or diffraction) explains each of the three observations: the echo from the cliff, the bending water waves at the bay, and the sound in the cove.</p> | <p>the wavelength of the wave. (b) Thunder has a low frequency, so by <math>\lambda = v/f</math> it has a long wavelength. This wavelength is comparable to the size of the headland, so the sound diffracts and spreads around it into the cove, even with no straight-line path. In the doorway experiment, 'Habari za asubuhi!' was heard around the corner for the same reason. (c) The microwaves of the phone signal diffract around the hill, but only imperfectly, so the signal is weak or fades in the valley behind. Engineers place repeater towers on hilltops to boost the signal into the valleys. (d) The fisherman hears thunder echoing from the cliff because of reflection. The water waves change direction at the shallow bay because of refraction: they slow down and bend towards the normal. The sound reaches the sheltered cove because of diffraction around the headland. All three are linked by the wave equation <math>v = f\lambda</math>.</p> |
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### Part 5: Applications, environment and the Scientist's Statement

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| <p>(a) Choose TWO Kenyan applications of wave properties from Lesson 6 (for example KMFRl sonar on Lake Victoria, ultrasound at Kenyatta National Hospital, cats-eye reflectors, or seismic monitoring by the Kenya Meteorological Department). For each, name the wave property it uses and explain how. (b) Describe ONE harmful effect of waves on the Kenyan environment and the wave property that causes it. (c) Describe ONE mitigation measure and explain the wave physics of how it works. (d) Write a Scientist's Statement of 4 to 6 sentences to the Kisumu fisherman explaining how wave science helps keep coastal communities safer.</p> | <p>(a) Sonar by KMFRl on Lake Victoria uses reflection: a pulse is sent down, the echo returns from the lake bed, and the time gives the depth using <math>d = (v \times t) / 2</math>. Ultrasound at Kenyatta National Hospital also uses reflection: sound pulses reflect from body tissues of different density and the returning echoes build an image. (Seismic monitoring uses refraction and diffraction as waves pass through and spread around rock layers; cats-eye reflectors reflect light back toward its source.) (b) Coastal erosion at Mombasa and Lamu: high-amplitude storm waves carry a lot of energy, and refraction over the changing water depth can focus wave energy onto headlands and unprotected shorelines, wearing away coral-stone foundations. (c) Sea walls reflect or absorb wave energy before it reaches the shoreline, so less energy reaches the land. Mangrove restoration also reduces the energy of the waves. (d) Dear fisherman, the thunder reached you around the headland because long-wavelength sound waves diffract. The water waves changed direction near the shore because they slow down in shallow water, which is refraction, and the echo came from the cliff by reflection. Using <math>v = f\lambda</math>, scientists can calculate how waves behave. Sonar and early-warning systems use this to help fishermen and coastal communities stay safe. Mangroves and sea walls reduce the energy of dangerous waves reaching your village.</p> |
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| RUBRIC                                                |                                                                                                                                                                                                                            |                                                                                                                                       |                                                                                                          |
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| Criterion                                             | Excellent (4)                                                                                                                                                                                                              | Proficient (3)                                                                                                                        | Developing (1–2)                                                                                         |
| Use of the wave equation and calculations             | All calculations are set out correctly with formulas, steps and units (4 Hz, 0.25 s, 150 m, 0.50 m and 0.47 m), and the inverse relationship between $f$ and $\lambda$ is clearly explained.                               | Most calculations are correct with some units or steps missing; the $f$ and $\lambda$ relationship is stated but not fully explained. | Calculations contain errors or are missing; the wave equation is stated without being applied correctly. |
| Explanation of reflection, refraction and diffraction | Each behaviour is defined accurately and the mechanism is explained (angle law; speed change at a boundary with bending towards the normal and constant frequency; diffraction greatest when size is close to wavelength). | Each behaviour is defined correctly but one mechanism is incomplete or imprecise.                                                     | Behaviours are confused with each other or described without any mechanism.                              |
| Connection to the anchoring phenomenon                | Each of the three observations (cliff echo, bending water waves, sound in the cove) is matched to the correct wave behaviour, and the limits of reflection and refraction are explained.                                   | Observations are matched to wave behaviours with minor gaps in the reasoning.                                                         | Few links are made to the fisherman at Kisumu, or the matches are incorrect.                             |
| Kenyan applications and environmental impact          | Applications, impacts and mitigation are accurate, use Kenyan contexts, and name the wave property and the physics of how it works.                                                                                        | Applications and impacts are correct but the physics of how they work is only partly explained.                                       | Examples are vague, not linked to wave properties, or incorrect.                                         |
| Scientific communication and Scientist's Statement    | Writing is clear, uses correct vocabulary, and the statement to the fisherman is accurate, plain-language and links wave science to community safety.                                                                      | Writing is mostly clear with some vocabulary slips; the statement is accurate but links to safety are brief.                          | Writing is unclear or vocabulary is misused; the statement is missing or inaccurate.                     |

## ATTRIBUTION AND LICENSING

**Curriculum Alignment:** Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

**Content & Pedagogy:** Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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